

CALORIE TRACKER



***SUBJECT: PROGRAMMING FOR PROBLEM
SOLVING USING PYTHON***

COURSE CODE: ETCCPP102

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CLASS: B.TECH CSE (AI ML) SECTION- B

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INTRODUCTION

The **Daily Calorie Tracker Application** is designed to help users monitor their **daily calorie intake** and expenditure with ease. In today's health-conscious world, keeping track of calories is essential for maintaining a **balanced diet** and **achieving fitness goals**. This application allows users to **log multiple meals**, **calculate total and average calorie intake**, and **compare it with a personalized daily goal**. It provides feedback on whether the user has met their calorie target and offers the option to save the meal data for future reference. The tool is a simple yet valuable utility for anyone interested in managing their diet effectively.

OBJECTIVE

- To enable users to **enter their calorie intake** for each meal consumed in a day.
- To calculate **total and average calories consumed** based on user input.
- To **compare consumed calories** against a set **daily calorie goal** and provide feedback.
- To **save daily meal records** with **timestamps** for tracking progress over time.
- To help users **build awareness** of their eating habits and **support long-term healthy lifestyle changes**.
- To provide a **simple, user-friendly** interface for beginners to practice Python programming concepts like **loops**, **conditionals**, **lists**, and **file handling**.
- To create a basic foundation that can be extended into more **advanced health or fitness tracking** applications in the future.

WORKING DESCRIPTION

The **Calorie Tracker Application** starts by welcoming the user and defining its working to them, which is to help in tracking user's **daily calorie intake and expenditure**. It first asks user to **add on the number of meals** they want to add on for the day, this input will control how many times loop will run for calorie intake per meal and store that inside an empty tuple further inside of list, for e.g. [(Breakfast, 150Kcal)].

Next, the program extracts all calorie values from the list of tuples and **calculates the total calories** consumed by summing them. It also **calculates the average calorie** intake per meal by dividing the total by the number of meals logged. Then, the program asks the user to **enter their daily calorie goal**, which serves as a benchmark for evaluating their intake. Using conditional statements, it compares the **total calories consumed to the daily goal**: if total **intake exceeds** the goal, it **warns the user** by how much; if it is below the goal, it congratulates the user and shows the remaining calories allowed; and if it matches exactly, it affirms the achievement.

After that, the program prints a **summary table** displaying **each meal name** alongside its **calorie content**, followed by the **total and average calories consumed**. Finally, the user is asked whether they want to save their meal data. If the user agrees, the program writes the **meal information, totals, averages**, and the **current date and time** into a text file named "**calorie_log.txt**." This feature provides persistent storage for users to review their calorie intake history later. The program then closes the file and confirms the save operation to the user.

This structured sequence guides the user through **meal logging, calorie tracking, goal comparison**, and data saving in a clear and interactive way, making the application practical for **everyday dietary management**. This detailed workflow explanation covers how data is **input, processed, compared, displayed**, and **optionally saved** for future reference.

Output Screenshots

- Output with warning message display

```
• Welcome to the Daily Calorie Tracker Application!
  This application helps you track your daily calorie intake and expenditure.

How many meals they would like to log today? 3
Enter the meal name (e.g., Breakfast, Lunch, Dinner, Snack): Breakfast
Enter the calories consumed for Breakfast in kcal: 50
Enter the meal name (e.g., Breakfast, Lunch, Dinner, Snack): Lunch
Enter the calories consumed for Lunch in kcal: 60
Enter the meal name (e.g., Breakfast, Lunch, Dinner, Snack): Dinner
Enter the calories consumed for Dinner in kcal: 45
Enter your daily calorie goal: 150
Warning: You have exceeded your daily calorie goal by 5.0 kcal.

Summary of your meals today:
Meal Name      Calories (kcal)
-----
Breakfast      50.0
Lunch          60.0
Dinner         45.0

Total: 155.0 kcal

Average: 51.666666666666664 kcal

Thank you for using the Daily Calorie Tracker Application! Stay healthy!
Would you like to save your meal data?
Enter 'yes' to save or 'no' to exit: yes
Your meal data has been saved to 'calorie_log.txt'.
```

- Saved Data

```
≡ calorie_log.txt X
≡ calorie_log.txt
1 Meal Name      Calories (kcal)
2 -----
3 Breakfast      50.0
4 Lunch          60.0
5 Dinner         45.0
6
7 Total Calories Consumed:      155.0 kcal
8
9 Average Calories per Meal:    51.666666666666664 kcal
10
11 Log Date and Time: 2025-12-02 13:41:11.259675
12
```

- Output congratulating when the calorie limit isn't broken

```
● Welcome to the Daily Calorie Tracker Application!
This application helps you track your daily calorie intake and expenditure.

How many meals they would like to log today? 4
Enter the meal name (e.g., Breakfast, Lunch, Dinner, Snack): Breakfast
Enter the calories consumed for Breakfast in kcal: 45
Enter the meal name (e.g., Breakfast, Lunch, Dinner, Snack): Lunch
Enter the calories consumed for Lunch in kcal: 60
Enter the meal name (e.g., Breakfast, Lunch, Dinner, Snack): Snacks
Enter the calories consumed for Snacks in kcal: 25
Enter the meal name (e.g., Breakfast, Lunch, Dinner, Snack): Dinner
Enter the calories consumed for Dinner in kcal: 45
Enter your daily calorie goal: 200
Good job! You are within your daily calorie goal. You have 25.0 kcal remaining.

Summary of your meals today:
Meal Name      Calories (kcal)
-----
Breakfast      45.0
Lunch          60.0
Snacks         25.0
Dinner         45.0

Total: 175.0 kcal

Average: 43.75 kcal

Thank you for using the Daily Calorie Tracker Application! Stay healthy!
Would you like to save your meal data?
Enter 'yes' to save or 'no' to exit: yes
Your meal data has been saved to 'calorie_log.txt'.
```

- Saved Data

```
≡ calorie_log.txt X
≡ calorie_log.txt
1 Meal Name Calories (kcal)
2 -----
3 Breakfast 45.0
4 Lunch 60.0
5 Snacks 25.0
6 Dinner 45.0
7
8 Total Calories Consumed: 175.0 kcal
9
10 Average Calories per Meal: 43.75 kcal
11
12 Log Date and Time: 2025-12-02 13:44:37.156493
13
```

Github Link : https://github.com/rsanshuman01-web-dev/Python_sem-1/tree/f08a6d44d9dcca0d920a37fb6c304ffb5e0a6a85/daily_calorie_tracker

REFLECTION

By making the **Daily Calorie Tracker Application**, I understood Python concepts like **loops**, **conditionals**, **lists**, and **file handling** in a simple and practical way. It gave me confidence in taking user **input**, **processing it**, and showing clear results to the user. This project also made me realise how small console programs can later be improved into full health or fitness tracking applications

The project also highlighted the importance of designing user-friendly interactions, such as **clear prompts**, **meaningful feedback messages**, and a simple summary of results to make the application easy to use. Adding the option to save data in a text file reinforced how persistence can make an application more useful by allowing users to track their habits over time.

CONCLUSION

The **Daily Calorie Tracker Application** provides a straightforward and efficient means for users to monitor and manage their calorie intake. By offering detailed **meal logging, calorie calculations, goal comparisons, and data saving functionality**, it supports users in maintaining healthier eating habits. This tool encourages awareness and control over daily nutrition, which is crucial for fitness and general well-being. Its simplicity and interactive design make it accessible for everyday use.
